Lily Houtman

cartographer and UI/UX designer

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pronouns: they/them

RESEARCH INTERESTS

mobile/digital cartography, accessibility, digital humanities, critical geography, UI/UX design, data journalism

EDUCATION

The Pennsylvania State University — 2024-present
Ph.D. Geography — ABD since May 2025
Advisor: Anthony C. Robinson
Committee: Cynthia Brewer, Mark Ortiz, Pamela VanHaitsma
Dissertation: *Evaluating the Production and Use of Mobile and Responsive Thematic Maps* (Forthcoming)

The Pennsylvania State University — 2022-2024
M.S. Geography
Advisor: Anthony C. Robinson
Committee: Cynthia Brewer and Helen Greatrex
Thesis: *Mobile Thematic Map Design in Data Journalism: An Evaluation of Techniques, Challenges, and Opportunities*

University of Wisconsin – Madison — 2018-2022
B.S. Cartography and GIS (Honors)
Advisor: Robert E. Roth
Thesis: *Mapping Presidential Elections: An Empirical Analysis of Effective Design Techniques*
B.S. Geography
B.S. Anthropology
Certificate in Archaeology

PROFESSIONAL and RESEARCH EXPERIENCE

Graduate Researcher – The Political-Industrial Ecology of Petrochemicals — 2025-present
The Pennsylvania State University, Department of Geography
PI: Dr. Jenn Baka

Developed mockups and assisted with the UI/UX design of *InFracStructure*, a tool which visualizes fracking and petrochemical resources in Pennsylvania to inform community members about the fracking-related emissions in their community. Led two user-centered design studies related to the tool, including preliminary interviews with NGO leaders. Developed in-person user evaluations with retired community members on mobile and desktop devices (planned date: May 2026). Will analyze results to understand how this complex interactive map works for the targeted users of the tool, and improve its design for future iterations.

Graduate Researcher – Non-Binary Maps to Reimagine Borderlands — 2024-present
The Pennsylvania State University, Department of Geography
PI: Dr. Melissa W. Wright

Collaborate with Dr. Melissa Wright to develop a cartographic and theoretical project that seeks to "unmap" the Rio Grande River Valley region and the borderlands of the United States and Mexico. Develop maps that consider the Rio Grande region from the constantly moving perspective of the watershed, instead of nation-state borders. Use creative cartographic design to highlight how existing infrastructure, including rail, river, and road network can defy and reimagine borders. Present maps in public venues, including classrooms and conferences. Contribute to academic work in the geohumanities through academic publications.

Graduate Student Cartographer – The GeoGraphics Lab 2023-present
The Pennsylvania State University, Department of Geography
Creative Director: Dr. Alicia Iverson

Apply six years of cartographic design experience to a variety of solo and group mapping projects in the GeoGraphics Lab, primarily through client projects. Work directly with clients including Penn State faculty, community partners, and external groups, to meet the unique design needs of each client. Work on projects related both to web mapping and physical map production, including maps for a large education center in the local community, Millbrook Marsh. As the most senior graduate student in the lab, serve as a mentor to graduate and undergraduate students, both for cartographic design and cartographic research.

Student Researcher – *Mapping for a Sustainable World*: QGIS Technical Supplement 2021-present
University of Wisconsin – Madison Cartography Lab, Department of Geography
Advisor: Dr. Robert Roth

Design a set of open-source QGIS technical mapping tutorials in 2021 and 2022 to support instruction about cartographic design. Directly supplement the joint International Cartographic Association and United Nations publication *Mapping for a Sustainable World* to support free and open education, data, and software. In 2025, implement updates to the tutorials based on feedback from multiple international workshops and feedback from educators who used the tutorials in classroom settings.

Graduate Research Assistant – Beescape NexGen Geovisualization Tool 2022-2025
The Pennsylvania State University, Departments of Geography and Entomology
PIs: Dr. Christina Grozinger and Dr. Anthony Robinson

Develop mockups based on previous focus groups for the tool *Beescape*, a decision support tool to assess the quality of the landscapes for supporting pollinators. Designed a printable version of the tool to assist field-based, offline use. Develop and implement an online user evaluation of beekeepers to assess the usability and utility of the web tool. Present the tool and related user evaluations in academic and public settings. Incorporate feedback from the most recent user test to develop a preliminary mobile version of the tool, also incorporating expert knowledge from my graduate research on mobile map design.

Graduate Student Collaborator – Harm Mapping: Visualizing Sexual Harassment 2023-2024
The Pennsylvania State University and Lafayette College
PI: Dr. Dana Cuomo

Collaborate with researchers at Lafayette College to take collected they collected on student experiences of sexual harassment, and incorporate it into maps to communicate the results with a broader audience. Consider the sensitive nature of the data and incorporate feminist principles to develop maps that will both serve the community and respect data privacy. Display maps in public settings on Lafayette College's campus to educate students and faculty about gender-based

violence on campus. Supply maps to project PI, who used the maps to argue to campus administration to implement changes in the campus built environment to better protect students.

Web Development Intern 2021-2022
United States Geological Survey, Web Mapping and Informatics Team

Update many existing science communication tools for the USGS, improving public outreach. Ensure all tools comply with accessibility standards for websites. Using JavaScript, added layers and improved the design of the Coastal Barrier Resource System Mapper, while staying within organizational design constraints. Using HTML, CSS, and Javascript, updated the Water Quality Portal, a database tool where users can access water quality data. Used Vue and Leaflet to begin the Real-Time Flood Impact Map, and documented the development process to pass the work on to another intern.

Undergraduate Student Cartographer – The UW Cartography Lab 2020-2022
University of Wisconsin – Madison, Department of Geography
Creative Director: Tanya Anderson

Work on a variety of client projects for community members and faculty affiliated with the university. Take on long term projects that require the management of a consistent style guide, including a series of 15 maps in grayscale for a history book, at times working with historical borders and historical data. Update the lab website to maintain current projects and improve visibility of current students. Collaborate with other students on larger group projects, including web-based mapping projects. Assist other students as needed, including a lead role in printing physical maps via a large plotter, and serving as technical support for plotter issues.

Undergraduate Curation Assistant – Biological Anthropology Teaching Collection 2019-2020
University of Wisconsin – Madison, Department of Anthropology
Museum Director: Elizabeth Leith

Assist in updating the Anthropology department teaching collection, to ensure consistency between physical and digital records. Transition records to digital systems as the university moves to primarily digital formats. Update physical labels on older collection items where records have worn off. Use expertise to identify unlabeled or miscatalogued collections. Record duplicate materials, such as cast replicas of hominin bones. Provide care to bones of modern humans, and identify any specimen which should not be in university ownership due to NAGPRA.

Museum Collections Intern – History and Natural History Summer 2019
Neville Public Museum
Collections Manager: Louise Pfotenhauer

Transition museum collections from closed exhibits to temporary and permanent storage, using physical and digital records to track locations. Select items from museum collections for public display in new exhibits, and draft complimentary descriptions of displayed items. With other museum staff, workshop ideas for new exhibits and explore collections for relevant materials, particularly with sensitivity to the depiction of marginalized groups. During this time, the museum's permanent display was undergoing a major transition, so another role was to talk with the public about this ongoing change.

PEER-REVIEWED PUBLICATIONS

Roth, RE, A Iverson, A Kagawa, G Le Sourd, B Ricker, T Andersen, G Baldrica-Franklin, **L Houtman**, F Ilunga, M Lee, M-J Kraak. 2025. Making and remaking Mapping for a Sustainable World: a

retrospective and prospective analysis as a case study in Processual Cartography. *International Journal of Cartography,* p. 1-40.

Houtman, L, AC Robinson, D McLaughlin, and C Grozinger. 2025. Evaluating the Usability and Utility of a Spatial Decision Support System for Pollinator Ecology. *Ecological Informatics, 89,* p. 1-12.

Houtman, L. 2025. Techniques, Challenges, and Opportunities in Mobile Thematic Map Design for Data Journalism. *Cartography and Geographic Information Science,* p. 1-20.

Houtman, L, D Cuomo, M Dennehy, M Forman, A Zea, and S Hannan. 2025. The Harm Mapping Project: Navigating Ethics and Collaboration in Map Design. *Cartographic Perspectives*, *105,* p. 27-33.

Song Z, RE Roth, **L Houtman**, T Prestby, A Iverson, and S Gao. 2023. Visual storytelling with maps: An empirical study on story map themes and narrative elements, visual storytelling genres and tropes, and individual audience differences. *Cartographic Perspectives, 100,* pp. 10-44.

BOOK and TUTORIAL PUBLICATIONS

Houtman L, G Baldrica-Franklin, and R.E. Roth. 2025. *Mapping for a Sustainable World: QGIS Technical Supplement*. Version 2.0 The University of Wisconsin Cartography Lab: Madison, WI. DOI: 10.5281/zenodo.5585647

Roth RE, G Gartner, G Baldrica-Franklin, A Cowart, **L Houtman**, A Kagawa, M Keskin, MJ Kraak, NT Nawshin, F Ortag, B Ricker, S Schlumpf, and Z Stachoň (Editors). 2025. *Atlas of Sustainability.* Vienna, Austria: The International Cartographic Association.

Houtman L, and RE Roth, 2021. *Mapping for a Sustainable World: QGIS technical supplement.* Version 1.0 The University of Wisconsin Cartography Lab: Madison, WI. DOI: 10.5281/zenodo.5585647

Roth RE, CM Sack, G Baldrica-Franklin, Y Chen, R Donohue, **L Houtman**, T Prestby, R Tolochko, and N Underwood, 2020. *Web Mapping: A Workbook for Interactive Cartography and Visualization on the Open Web*. Version 1.0. University of Wisconsin Cartography Laboratory: Madison, WI. DOI: 10.5281/zenodo.4685763

CONFERENCE PUBLICATIONS

Houtman L, G Baldrica-Franklin, and RE Roth. 2026. Reintroducing a QGIS Technical Supplement for *Mapping for a Sustainable World*. *Abstracts of the ICA (Cartography and Sustainable Development Workshop) (accepted)*

Houtman, L. 2025. Characterizing the current state of mobile thematic cartography. *Abstracts of the ICA (International Cartographic Conference)*, *10*, p. 111.

Houtman, L. 2024. Future opportunities for mobile thematic map design in data journalism. *Abstracts of the ICA (European Cartographic Conference), 7*, p. 59.

Houtman, L. 2023. At the Intersection of Mobile Thematic Cartography and Data Journalism. *Abstracts of the ICA (International Cartographic Conference), 6*, p. 93.

Robinson AC, TJ Prestby, D McLaughlin, **L Houtman**, P Dudas, and C Grozinger. 2023. Geovisualization for Pollinator Decision Support with Beescape NexGen. *Abstracts of the ICA*

(International Cartographic Conference), 6, p. 212.

Houtman L. 2022. Mapping US Elections: An Empirical Analysis of Design Techniques. *Abstracts of the ICA (European Cartographic Conference), 5*, p. 126.

Houtman L. and RE Roth. 2021. Introducing a QGIS Technical Supplement for *Mapping for a Sustainable World*. *Abstracts of the ICA (International Cartographic Conference), 3,* p. 113.

Roth RE, CM Sack, G Baldrica-Franklin, Y Chen, R Donohue, **L Houtman**, T Prestby, R Tolochko, and N Underwood, 2021. Introducing Web Mapping: A Workbook for Interactive Cartography and Visualization on the Open Web. *Abstracts of the ICA (International Cartographic Conference)*, *3*, p. 254.

BOOK REVIEWS

Houtman, L. 2024. Review of The Art of Insight: How Great Visualization Designers Think. *Cartographic Perspectives, 104,* p. 70-73.

Houtman, L. 2024. Review of Airline Maps: A Century of Art and Design. *Cartographic Perspectives, 103,* p. 99-102.

INVITED TALKS and LECTURES

Invited Talk: "Techniques, challenges, and opportunities in mobile thematic map design for data journalism" Fall 2025
Department of Computing, Georgia Institute of Technology
Invited by Dr. Clio Andris for CS8803: Interactive Maps and Geovisualization

Guest Lecture: "Multivariate Maps" Fall 2025
Department of Geography, The Pennsylvania State University
Invited by Dr. Alicia Iverson for GEOG 361: Introduction to Cartography

Invited Talk: "Art, Activism, and Cartography" Fall 2025
Department of Sociology, American University
Invited by Dr. Felicia Henry for SOCY 358: Arts-Based Activism

Guest Lecture: "Multivariate Maps" Spring 2025
Department of Geography, The Pennsylvania State University
Invited by Tim Prestby for GEOG 361: Introduction to Cartography

Guest Lecture: "Election Mapping - America in Red & Blue" Fall 2023
Department of Geography, The Pennsylvania State University
Invited by Dr. Anthony Robinson for GEOG 6N: Maps and the Geospatial Revolution

Guest Lecture: "Maps and Power" Spring 2023
Department of Women, Gender, and Sexuality Studies, The Pennsylvania State University
Invited by Dr. Jill Wood for WMNST 197: Creating Safer Campuses

CONFERENCE PRESENTATIONS

Houtman, L. "Mobile Thematic Map Design: An Analysis of Young Adult Audiences Via Creative Focus Groups" *presented at* the American Association of Geographers Annual Meeting. March 2026. San Francisco, California.

Houtman, L. “Designing Non-Binary Maps to Reimagine Borderlands” *presented at* the 45th meeting of the North American Cartographic Information Society (NACIS). October 2025. Louisville, Kentucky.

Houtman, L. “Characterizing the current state of mobile thematic cartography” *presented at* the 32nd International Cartographic Conference (ICC). August 2025. Vancouver, Canada.

Houtman, L. “No Boundaries, No Borders: Creating Non-Binary Maps” *presented at* the No Boundaries Symposium. May 2025. State College, Pennsylvania.

Houtman, L., Wright, M.W. “An Exercise in Non-Binary Maps” *presented at* the American Association of Geographers Annual Meeting. March 2025. Detroit, Michigan.

Houtman, L. "The Challenges and Opportunities of Designing News Maps for Mobile Devices" *presented at* the Computation and Journalism Symposium. October 2024. Boston, Massachusetts.

Houtman, L. “Discussing Current Limits and Future Opportunities in Mobile Map Design” *presented at* the 44th meeting of the North American Cartographic Information Society (NACIS). October 2024. Tacoma, Washington.

Houtman, L. “Future Opportunities for Mobile Thematic Map Design in Data Journalism” *presented at* EuroCarto. September 2024. Vienna, Austria.

Houtman, L. “Using Beescape NexGen: A Pollinator Health Decision-Support Tool” *presented at* the Technologies for Agriculture and Living Systems Symposium. May 2024. State College, Pennsylvania.

Houtman, L. “Mobile Thematic Cartography: Interviews on Design and Practice” *presented at* the American Association of Geographers Annual Meeting. April 2024. Virtual.

Houtman, L. “How Should We Design Mobile Thematic Maps” *presented at* the 43rd meeting of the North American Cartographic Information Society (NACIS). October 2023. Pittsburgh, Pennsylvania.

Houtman, L.“At the Intersection of Mobile Thematic Cartography and Data Journalism” *presented at* the 31st International Cartographic Conference (ICC). August 2023. Cape Town, South Africa.

Houtman, L.“The Challenges of Using AI to Promote Further Learning in Cartographic Education” *presented at* the Pre-Conference Workshop on Cartography and AI at the 31st International Cartographic Conference (ICC). August 2023. Stellenbosch, South Africa.

Houtman, L., Prestby, T.J., McLaughlin, D., Robinson, A.C., & Grozinger, C.“Designing Beescape NexGen: A Geovisualization Tool to Support Pollinator Health” *presented at* the American Association of Geographers Annual Meeting. March 2023. Denver, Colorado.

Houtman, L. “How Should We Design Election Maps” *presented at* the 42nd meeting of the North American Cartographic Information Society (NACIS). October 2022. Minneapolis, Minnesota.

Houtman, L. “Mapping US Elections: An Empirical Analysis of Design Techniques” *presented at* EuroCarto. September 2022. Vienna, Austria.

Houtman, L. “Mapping Presidential Elections: An Empirical Analysis of Effective Design Techniques” *presented at* the American Association of Geographers (AAG) Annual Meeting. March 2022. Virtual.

Houtman, L., Roth R.E.“Introducing a QGIS technical supplement for Mapping for a Sustainable World” *presented at* the 30th International Cartographic Conference (ICC). December 2021. Florence, Italy.

PANEL DISCUSSIONS

Co-Moderator (*with* Meghan Kelly), "Cartographies of Interdependence: Intersectional Feminist Mapping in Community Struggles" *at* the 45th meeting of the North American Cartographic Information Society (NACIS). October 2025. Louisville, Kentucky.

Panelist, "LGBTQ+ in STEM Leading the Way" *at* The Pennsylvania State University. April 2025. State College, Pennsylvania.

Panelist, "The Future of University Cartography Labs 2: Student Experience and Creating New Spaces" *at* the American Association of Geographers Annual Meeting. April 2024. Virtual.

Panelist, "Diversity, Equity, and Inclusion II: Queer Cartography" *at* the 43rd meeting of the North American Cartographic Information Society (NACIS). October 2023. Pittsburgh, Pennsylvania.

Co-Moderator (*with* F.M. Kim), "Finding Queerness in GIS and Cartography" *at* the American Association of Geographers Annual Meeting. March 2023. Denver, Colorado.

ORGANIZED WORKSHOPS and SESSIONS

Co-Organizer (*with* Alicia Iverson and Felicia Henry), "Introduction to Mapping for Social Change Workshop" *at* American University. April 2026. Washington, D.C.

Co-Organizer (*with* Silas Hill), "DEI Speaker Series: Cartographies of Interdependence: Intersectional Feminist Mapping in Community Struggles" *at* the 45th meeting of the North American Cartographic Information Society (NACIS). October 2025. Louisville, Kentucky.

Co-Organizer (*with* Gareth Baldrica-Franklin), "Unmapping Workshop", NACIS Commons *at* the 45th meeting of the North American Cartographic Information Society (NACIS). October 2025. Louisville, Kentucky.

Co-Organizer (*with* Robert E. Roth, et al.), "T/U(X)W Design Challenge" *at* the Technical University of Vienna (TU Wien). May 2025. Virtual.

Co-Organizer (*with* Gareth Baldrica-Franklin), "Critical Cartographic Processes" *at* the American Association of Geographers Annual Meeting. March 2025. Detroit, Michigan.

Co-Organizer (*with* Gareth Baldrica-Franklin), "Unmapping: Creative and Critical Cartography for All", NACIS Commons *at* the 44th meeting of the North American Cartographic Information Society (NACIS). October 2024. Tacoma, Washington.

Co-Organizer (*with* Zhaoxu Sui), "Putting the Human back in Cartography: Designs and Practices" *at* the American Association of Geographers Annual Meeting. April 2024. Virtual.

Co-Organizer (*with* Gareth Baldrica-Franklin), "Unmapping: Creative and Critical Cartography for All", NACIS Commons *at* the 43rd meeting of the North American Cartographic Information Society (NACIS). October 2023. Pittsburgh, Pennsylvania.

Co-Organizer (*with* F.M. Kim), "Finding Queerness in GIS and Cartography" *at* the American Association of Geographers Annual Meeting. March 2023. Denver, Colorado.

POSTER PRESENTATIONS

Houtman, L. "The Challenges and Constraints of Designing Mobile News Maps for Science Communication" *presented at* the Viral Movements Symposium. May 2024. State College, Pennsylvania.

Houtman, L. "Evaluating the Use and Production of Mobile News Maps" *presented at* the Pennsylvania State University College of Earth and Mineral Sciences NSF GRFP Poster Session. October 2023. State College, Pennsylvania.

Houtman, L. "Mapping Presidential Elections: An Empirical Analysis of Effective Design Techniques" *presented at* the 41st meeting of the North American Cartographic Information Society (NACIS). October 2021. Oklahoma City, Oklahoma.

Houtman, L. "Human Evolution in the East African Rift Valley" *presented at* the 41st meeting of the North American Cartographic Information Society (NACIS). October 2021. Oklahoma City, Oklahoma.

TEACHING EXPERIENCE

Project Mentor and Instructor: Independent Study Spring 2026
Department of Geography, The Pennsylvania State University
GEOG 494 - Research Project in Geography

Curriculum Developer for Online Geospatial Education Fall 2024
Online Geospatial Education Program, The Pennsylvania State University
GEOG 486 - Cartography and Visualization (*with* Fritz Kessler and Marcela Suárez)

Undergraduate Teaching Intern: Interactive Cartography Fall 2021
Department of Geography, University of Wisconsin – Madison
Geography 575 - Interactive Cartography and Geovisualization (*with* Robert. E Roth & Gareth Baldrica-Franklin)

Undergraduate Teaching Intern: Introduction to Cartography Fall 2021
Department of Geography, University of Wisconsin – Madison
Geography 370 - Introduction to Cartography (*with* Robert E. Roth & Alicia Iverson)

FELLOWSHIPS and GRANTS

$13,000 **Interdisciplinary Approaches to Human-Environment Interaction** 2025-2026
The Pennsylvania State University Community, Science, and Society Initiative (CSSI)
Collaborator, "Mapping Against the Binary in the Mexico-US Borderlands"

$111,000 **Graduate Research Fellowship (GRFP)** 2024-2027
National Science Foundation (NSF)
Awardee, "Evaluating the Use and Production of Mobile News Maps"

$9,000 **Letters & Science Honors Mark Mensink Senior Honors Thesis Grant** 2021-2022
University of Wisconsin – Madison
Awardee, "Mapping Presidential Elections: An Empirical Analysis of Design Techniques"

$4,000 **McPherson Eye Research Institute Undergraduate Research Fellowship** 2021-2022
University of Wisconsin – Madison
Awardee, "Mapping Presidential Elections: An Empirical Analysis of Design Techniques"

HONORS, SCHOLARSHIPS, and TRAVEL SUPPORT

$700	**Abler and Erickson Graduate Student Enrichment Funding Award** *Penn State University, Department of Geography*	2026
N/A	**Charles Hosler Diversity, Equity, and Inclusion Graduate Student Award** *Penn State University, College of Earth and Mineral Sciences (EMS)*	2026
$500	**NACIS Student Travel Grant** *North American Cartographic Information Society*	2025
$1,800	**USNC Early Career Scholar Funding for Participation at ICC** *US National Committee for the ICA*	2025
$1,000	**ICA Young Scholars Funding for Participation at ICC** *Executive Committee of the ICA*	2025
$50	**No Boundaries Symposium Award for Best Paper (Third Place)** *Penn State University, Department of Geography*	2025
$1,500	**Penn State Graduate Student International Travel Grant** *Penn State Global*	2025
$500	**Myers Award for Excellence in Cartographic Design** *Penn State University, Department of Geography*	2025
$1,500	**Doctoral Student Scholarship Award** *Cartography and Geographic Information Society (CAGIS)*	2024
$300	**Master's Thesis Research Grant** *Cartography and Mapping Speciality Group of the AAG*	2024
$500	**NACIS Student Travel Grant** *North American Cartographic Information Society*	2024
$5,000	**Penn State EMS NSF GRFP Proposal Poster Competition** *Penn State College of Earth and Mineral Sciences*	2023
$500	**NACIS Student Travel Grant** *North American Cartographic Information Society*	2023
$200	**NACIS Diversity, Equity, and Inclusion Travel Grant** *North American Cartographic Information Society*	2023
$2,700	**USNC Early Career Scholar Funding for Participation at ICC** *US National Committee for the ICA*	2023
$600	**ICA Young Scholars Funding for Participation at ICC** *Executive Committee of the ICA*	2023
$5,000	**Fund for Excellence in Graduate Recruitment (FEGR)** *The Pennsylvania State University*	2022-2023
$500	**NACIS Student Travel Grant** *North American Cartographic Information Society*	2022
$100	**Excellence in the Department of Geography** *University of Wisconsin – Madison*	2022
$100	**Achievement in in the Cartography/GIS Major** *University of Wisconsin – Madison*	2022
$2,000	**USNC Early Career Scholar Funding for Participation at ICC** *US National Committee for the ICA*	2021
$50	**Barbara Bartz Petchenik Award for Cartographic Design (First Place)** *University of Wisconsin – Madison*	2021
$1,000	**Department of Geography Kohn Award**	2021

University of Wisconsin – Madison

$50 **Barbara Bartz Petchenik Award for Cartographic Design (Second Place)** 2020
University of Wisconsin – Madison

SERVICE and LEADERSHIP

Chair, NACIS Shop Initiative 2025-present
North American Cartographic Information Society (NACIS)

Co-Moderator (with Gareth Baldrica-Franklin), Practical Cartography Day 2025
North American Cartographic Information Society (NACIS)

NACIS Board Director-At-Large 2024-present
North American Cartographic Information Society (NACIS)

NSF GRFP Workshop Peer Mentor 2024-present
College of Earth and Mineral Sciences, The Pennsylvania State University

Coffee Hour (Colloquium) Graduate Student Committee Member 2024-present
Department of Geography, The Pennsylvania State University

Co-Moderator (with Nat Douglass), Practical Cartography Day 2024
North American Cartographic Information Society (NACIS)

Justice, Equity, Diversity, and Inclusion (JEDI) Student Representative 2023-present
American Association of Geographers (AAG)

Supporting Women in Geography (SWIG) President 2023-present
Department of Geography, The Pennsylvania State University

Graduate Student Representative 2023-2024
Department of Geography, The Pennsylvania State University

Supporting Women in Geography (SWIG) Board Member 2022-2023
Department of Geography, The Pennsylvania State University

Social Media Coordinator and Board Member 2021-2022
Geography Club, University of Wisconsin-Madison

External Relations Committee Undergraduate Student Representative 2021-2022
Department of Geography, University of Wisconsin-Madison

PEER REVIEW

GeoHumanities 2025-present
International Journal of Geographical Information Science 2024-present
Cartographic Perspectives 2024-present
Cartography and Geographic Information Science (CaGIS) 2024-present

TECHNICAL SKILLS

Web Development and Cartography

- JavaScript *(including Leaflet and D3 map libraries)*
- HTML/CSS/CSS
- FrontEnd Web Frameworks *(Bootstrap, Vue)*
- QGIS
- Esri Suite *(ArcGIS JS API, ArcGIS Pro, AGOL, StoryMaps)*
- Git/GitHub

Data Visualization and Analysis

- Inkscape
- Adobe Creative Cloud *(Illustrator, Photoshop, InDesign)*
- R and RStudio
- Tableau
- Blender
- Microsoft Excel

MEMBERSHIPS

Cartography and Geographic Information Society (CaGIS)	2024-present
American Association of Geographers (AAG)	2021-present
International Cartographic Association (ICA)	2021-present
The Phi Beta Kappa Society (PBK)	2021-present
North American Cartographic Information Society (NACIS)	2020-present